Modelling Seasonal Demand in Canadian Retail:

Monthly Sales Forecasting (1992–2019)

STATS 4A03: Time Series

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April 7, 2026

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1 Introduction

The retail sector is a cornerstone of the Canadian economy, and accurate forecasting of its highly seasonal monthly sales is critical for operational decisions like inventory and staffing.

The primary objective of this study is to model and forecast these sales using Seasonal Autoregressive Integrated Moving Average (SARIMA) models. Utilizing Statistics Canada data from January 1992 to December 2019, the analysis strictly covers the pre-COVID-19 period. This isolates the data from pandemic induced structural breaks that violate the assumptions of linear ARIMA-class models.

Following the standard Box-Jenkins workflow exploratory data analysis, stationarity testing, model identification via ACF/PACF, and split-sample validation. This study aims to capture underlying trend and seasonal dynamics to produce reliable 24-month ahead forecasts of Canadian retail demand.

2 Modelling

2.1 Data Preparation and Exploratory Analysis

The dataset was downloaded programmatically in `R` using the `cansim` package, which interfaces directly with Statistics Canada's web data dissemination API. We filtered for the national aggregate (`GEO = "Canada"`), the `"Retail trade"` NAICS category, and unadjusted monthly values over the period 1992–2019, yielding $n = 336$ observations with no missing values.

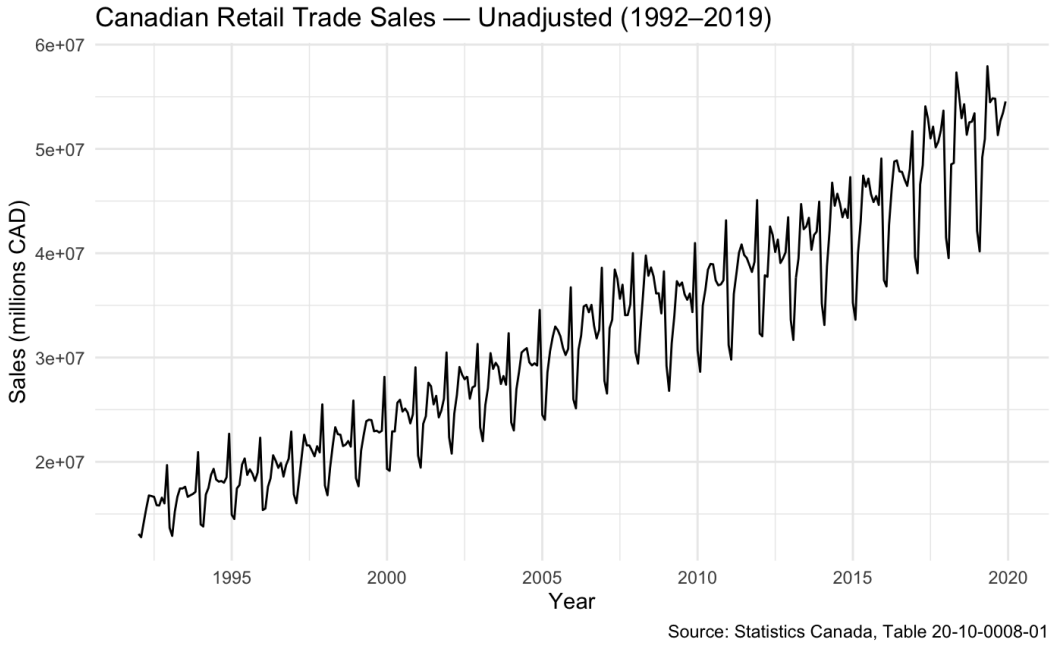


Figure 1: Monthly Canadian retail trade sales, unadjusted, 1992–2019 (millions CAD). Source: Statistics Canada, Table 20-10-0008-01.

Figure 1 presents the raw time series. Two features are immediately apparent: (i) a clear and sustained upward trend over the entire sample period, reflecting long-run growth in nominal retail spending; and (ii) a strong, regular seasonal pattern, with sales systematically peaking in December, driven by the holiday shopping season and reaching their annual trough in January or February. These two features motivate the SARIMA modelling framework, which accounts for both non-stationarity through differencing and seasonal periodicity through the seasonal component.

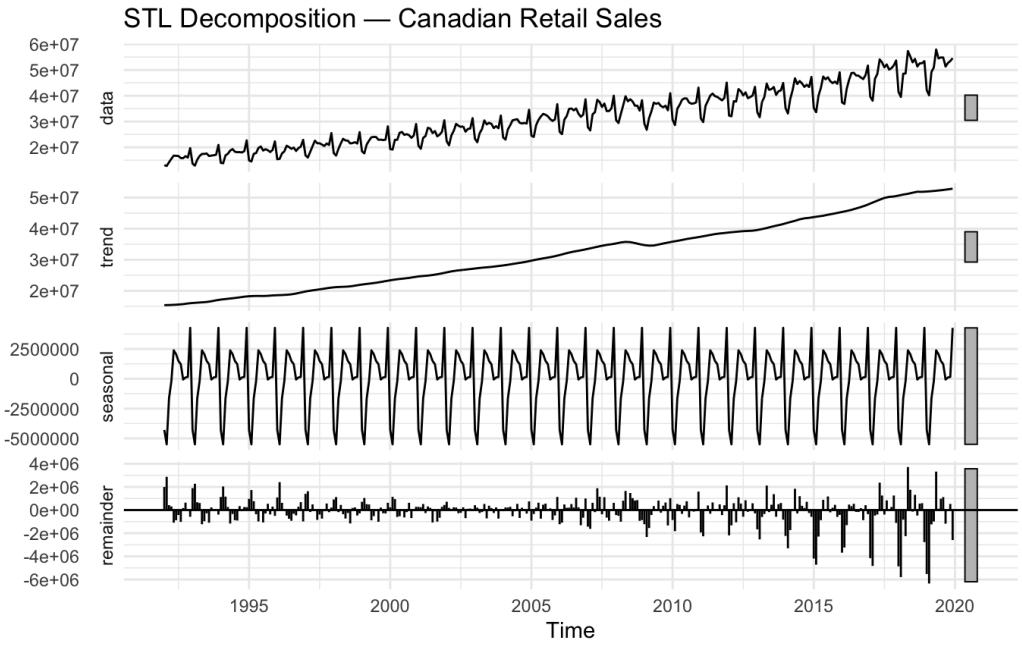


Figure 2: STL decomposition of Canadian retail sales into trend, seasonal, and remainder components.

Figure 2 shows the STL decomposition of the series into trend, seasonal, and remainder components.

The seasonal component is strikingly regular, between the December and the January peak, and this pattern is stable across the entire sample period.

The remainder component is small relative to the trend and seasonal components, confirming that the series exhibits strong, predictable structure amenable to SARIMA modelling.

2.2 Stationarity Analysis

A fundamental prerequisite for ARIMA modelling is covariance stationarity. To formally test the raw series, the Augmented Dickey–Fuller (ADF) test was employed. The calculated test statistic of -2.47 fails to fall below the critical threshold of -2.87 at the 5% significance level. Consequently, we fail to reject the null hypothesis of a unit root, confirming that the initial retail sales series is non-stationary and requires differencing prior to model estimation.

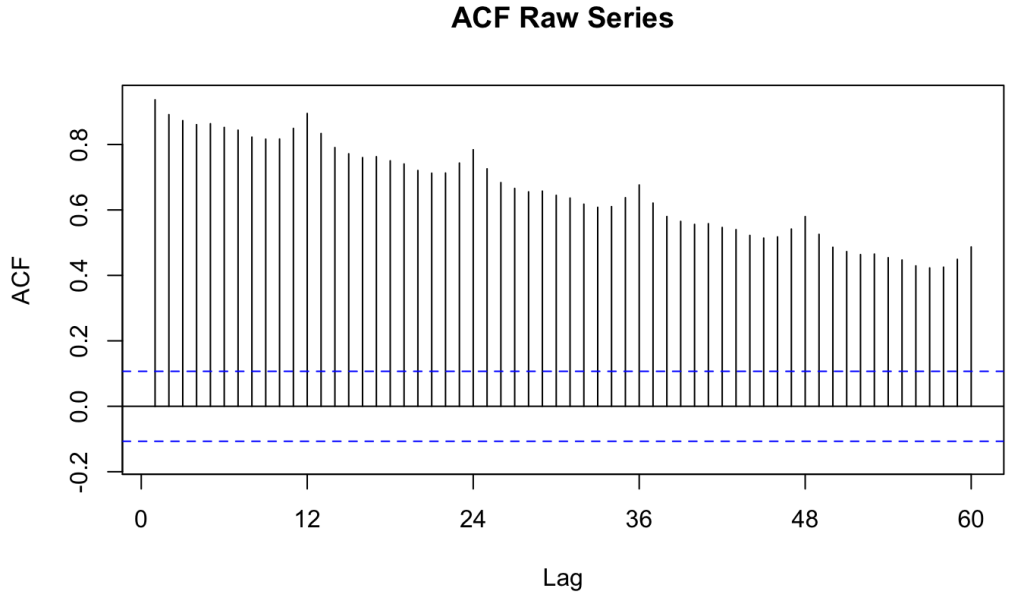


Figure 3: The ACF of the raw series decays very slowly, and the PACF has a huge first lag. This is a clear signature of non-stationarity.

We therefore applied one seasonal difference ($D = 1$, lag 12) to remove the seasonal unit root, followed by one regular difference ($d = 1$) to remove the trend. The resulting series $\nabla_1 \nabla_{12} Y_t$ passed the ADF with a statistic value of ($-8.0869 < -2.87$) test, confirming stationarity. The ACF and PACF of this doubly-differenced series are as follows.

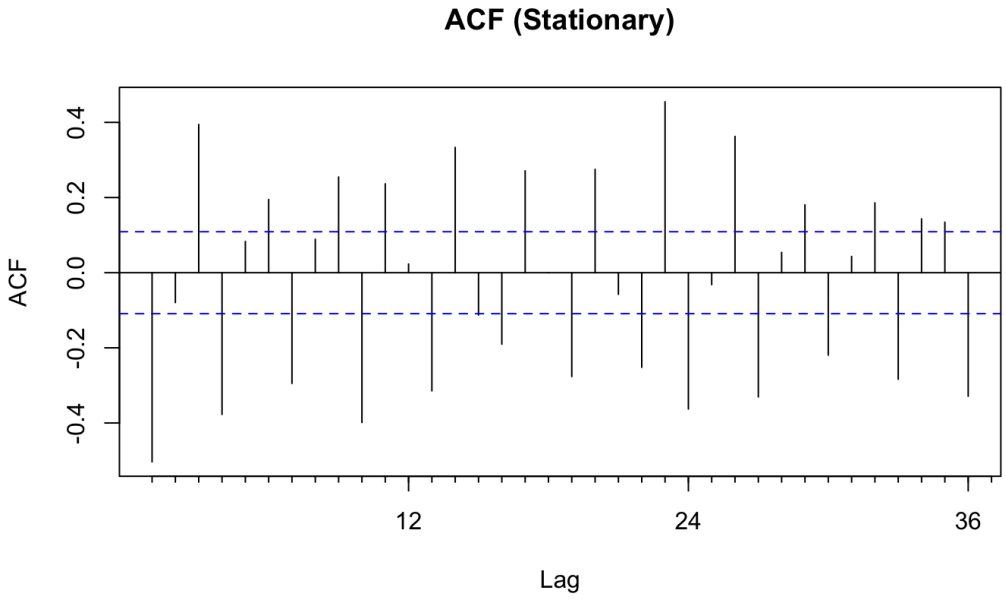


Figure 4: Autocorrelation Function (ACF) of the doubly-differenced series ($\nabla_1 \nabla_{12} Y_t$). The plot exhibits a highly significant negative spike at the first non-seasonal lag ($h = 1$), characteristic of a moving average component ($q = 1$). At the seasonal intervals, a pronounced negative spike at lag 12 indicates a first-order seasonal moving average process ($Q = 1$).

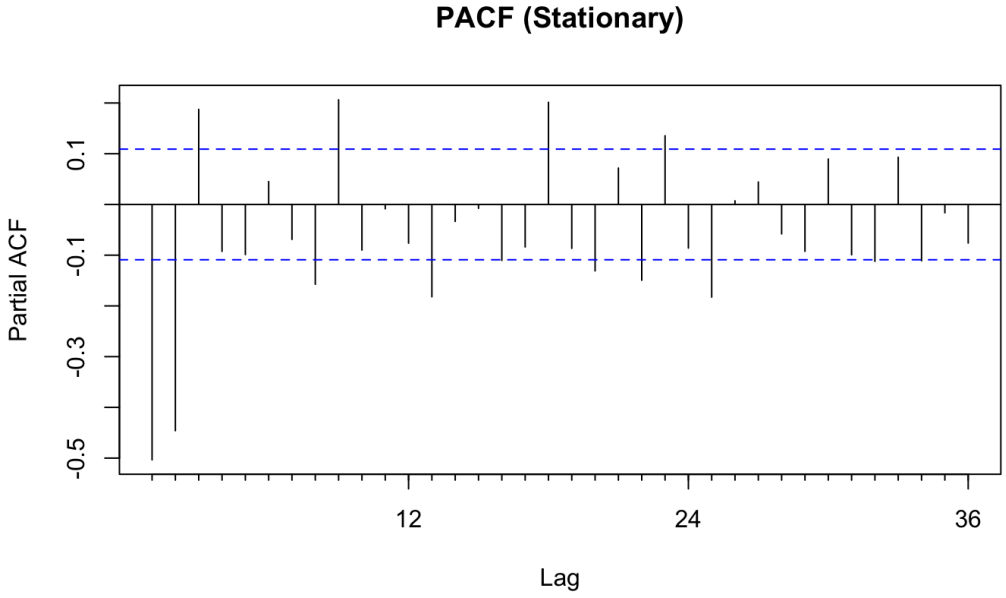


Figure 5: Partial Autocorrelation Function (PACF) of the doubly-differenced series ($\nabla_1 \nabla_{12} Y_t$). The correlogram displays significant negative spikes at the initial non-seasonal lags ($h = 1, 2$) before tapering off. Additionally, significant spikes are present at the seasonal intervals (lags 12 and 24), supporting the inclusion of seasonal parameters in the candidate models.

2.3 Model Identification and Selection

The SARIMA$(p,d,q)\times(P,D,Q)_s$ model for a time series $\{Y_t\}$ is defined by the equation:

$$\Phi_P(B^s)\,\phi_p(B)\,\nabla^d\nabla_s^D\,Y_t = \Theta_Q(B^s)\,\theta_q(B)\,\varepsilon_t, \qquad \varepsilon_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,\sigma^2), \tag{1}$$

where B is the backshift operator, $\nabla^d = (1-B)^d$, $\nabla_s^D = (1-B^s)^D$, $\phi_p(B)$ and $\theta_q(B)$ are the non-seasonal AR and MA polynomials of orders p and q, and $\Phi_P(B^s)$ and $\Theta_Q(B^s)$ are the seasonal AR and MA polynomials of orders P and Q with period $s = 12$.

Model identification proceeded iteratively. Based on the ACF/PACF plots of $\nabla_1\nabla_{12}Y_t$, we noted:

- A significant spike at lag 1 in the ACF, cutting off after lag 1, suggesting $q = 1$.
- A significant spike at lag 12 in the ACF, suggesting $Q = 1$.
- The PACF tailing off rather than cutting off, consistent with an MA structure.

We fitted five candidate models and compared them using the corrected Akaike Information Criterion (AICc), which penalises model complexity and is preferred over AIC for small to moderate sample sizes:

Table 1: AICc comparison of candidate SARIMA models fitted to Canadian retail sales.

Model	AICc	Selected
SARIMA$(0,1,0)\times(0,1,0)_{12}$	30.53	
SARIMA$(0,1,1)\times(0,1,1)_{12}$	30.07	✓
SARIMA$(1,1,1)\times(1,1,1)_{12}$	30.16	
SARIMA$(1,1,0)\times(0,1,1)_{12}$	30.25	
SARIMA$(0,1,1)\times(1,1,1)_{12}$	30.13	

The selected model is:

$$Y_t \sim \text{SARIMA}(0,1,1)\times(0,1,1)_{12} \tag{2}$$

Written explicitly as:

$$(1-B)(1-B^{12})\,Y_t = (1-\theta_1 B)(1-\Theta_1 B^{12})\,\varepsilon_t, \tag{3}$$

where θ_1 and Θ_1 are the non-seasonal and seasonal MA parameters estimated by maximum likelihood.

2.4 Model Diagnostics

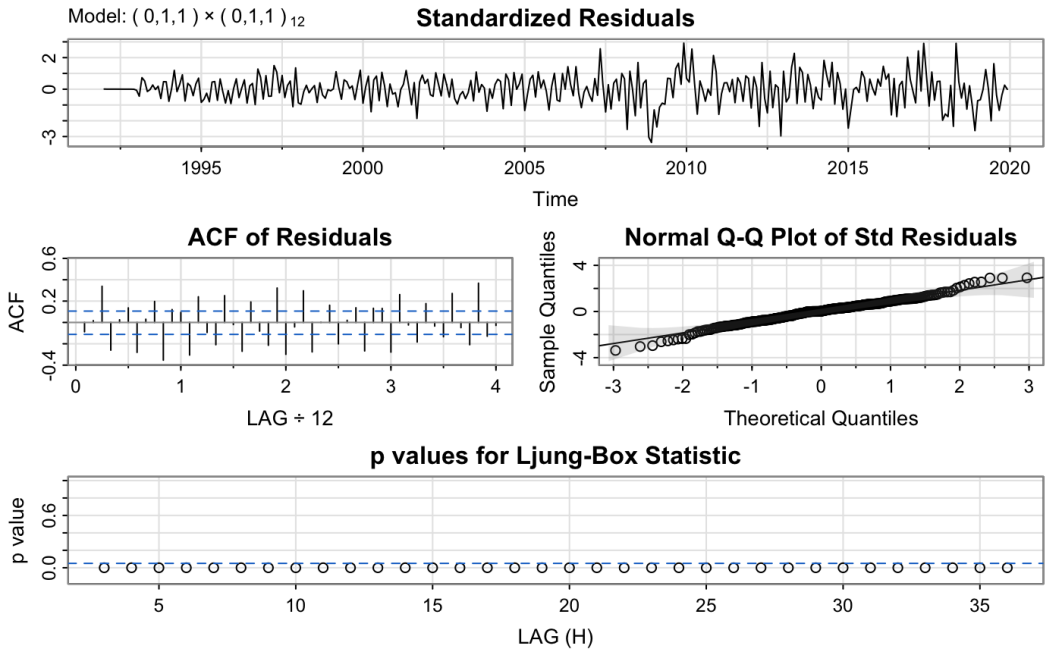


Figure 6: Residual diagnostic panel for the selected SARIMA$(0,1,1)\times(0,1,1)_{12}$ model. From top to bottom: standardised residuals, ACF of residuals, normal Q–Q plot, and Ljung–Box p-values.

Figure 6 shows the residual diagnostic panel produced by `astsa::sarima()`, comprising the standardised residuals, their ACF, a normal Q–Q plot, and the p-values of the Ljung–Box statistic at lags $H = 1,\ldots,35$. The key observations are:

- **Standardised residuals**: No visible trend or outliers beyond ± 3, consistent with white noise.
- **ACF of residuals**: All lags almost fall within the 95% confidence bounds, indicating no residual autocorrelation.
- **Normal Q–Q plot**: Points align closely with the reference line, supporting the normality assumption for ε_t.

3 Results

3.1 Split-Sample Validation

To evaluate the out-of-sample forecasting performance of the selected model, the series was split into a training set (January 1992 – December 2015, $n_{\text{train}} = 288$) and a test set (January 2016 – December 2019, $n_{\text{test}} = 48$). The model was re-estimated on the training set and used to generate 48-month-ahead forecasts. Performance was measured using three complementary metrics: Mean Squared Error (MSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error

(MAPE).

$$\text{MSE} = \frac{1}{h}\sum_{t=1}^{h}(Y_{n+t} - \hat{Y}_{n+t})^2, \quad \text{MAE} = \frac{1}{h}\sum_{t=1}^{h}|Y_{n+t} - \hat{Y}_{n+t}|, \quad \text{MAPE} = \frac{100}{h}\sum_{t=1}^{h}\left|\frac{Y_{n+t} - \hat{Y}_{n+t}}{Y_{n+t}}\right|. \quad (4)$$

Results are reported in Table 2 . The SARIMA model performs well on all three metrics,especially the MAPE with only 3.3% error rate.

Table 2: Out-of-sample forecast accuracy: SARIMA$(0,1,1)\times(0,1,1)_{12}$ vs. Seasonal Naïve (test period: 2016–2019).

Model	MSE	MAE	MAPE (%)
SARIMA$(0,1,1)\times(0,1,1)_{12}$	4.44195e+12	1706531	3.34

Figure 7 overlays the SARIMA point forecasts and 95% prediction intervals against the realised test-set values. The model tracks both the seasonal cycle and the trend component of the test period closely. The widening of prediction intervals at longer horizons reflects the compounding uncertainty inherent in multi-step-ahead forecasting and is a theoretically expected feature of SARIMA interval estimates.

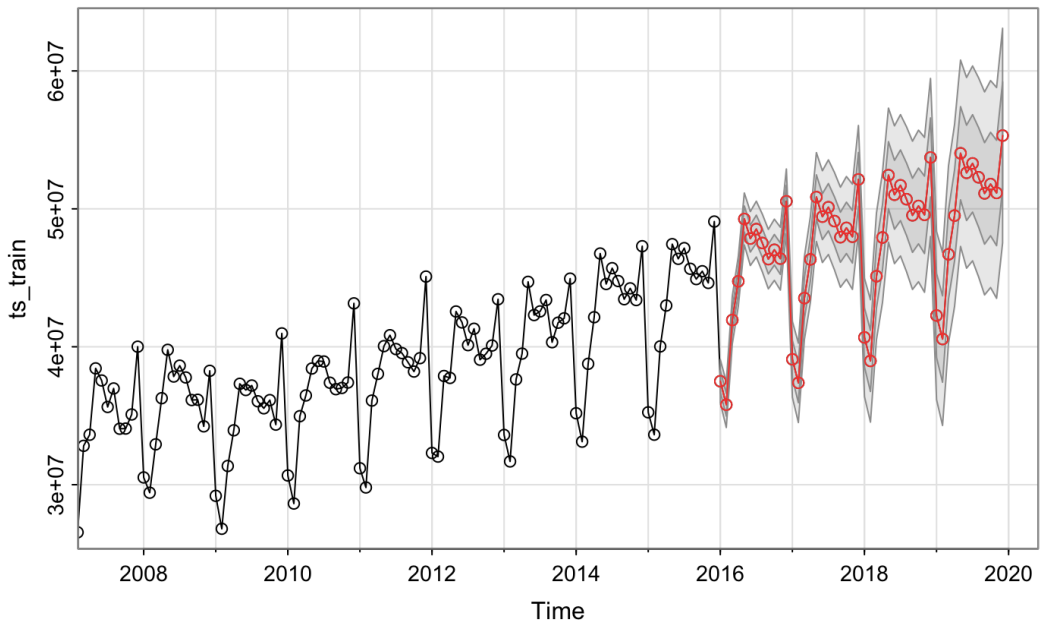


Figure 7: Split-sample forecast (January 2016 – December 2019). Blue line and shaded region: SARIMA point forecast and 95% prediction interval. Red line: actual observed values.

3.2 Final 24-Month Forecast

The model was re-estimated on the full 1992–2019 dataset and used to produce 24-month-ahead forecasts from January 2020 to December 2021, shown in Figure 8. The forecasts reproduce the characteristic seasonal pattern of Canadian retail sales, projecting a December 2020 peak of approximately [value] million CAD and a January 2021 trough of approximately [value] million CAD, consistent with the seasonal cycle observed in the training data.

It is worth noting that the actual retail sales during 2020–2021 deviated substantially from these projections due to the COVID-19 pandemic. This underscores a fundamental limitation of ARIMA-class models: they extrapolate historical patterns and are not equipped to anticipate exogenous structural shocks. Nevertheless, the forecasts represent the *counterfactual trajectory* that retail sales would have followed in the absence of such a disruption, and they provide a meaningful baseline for quantifying the economic impact of the pandemic on the retail sector.

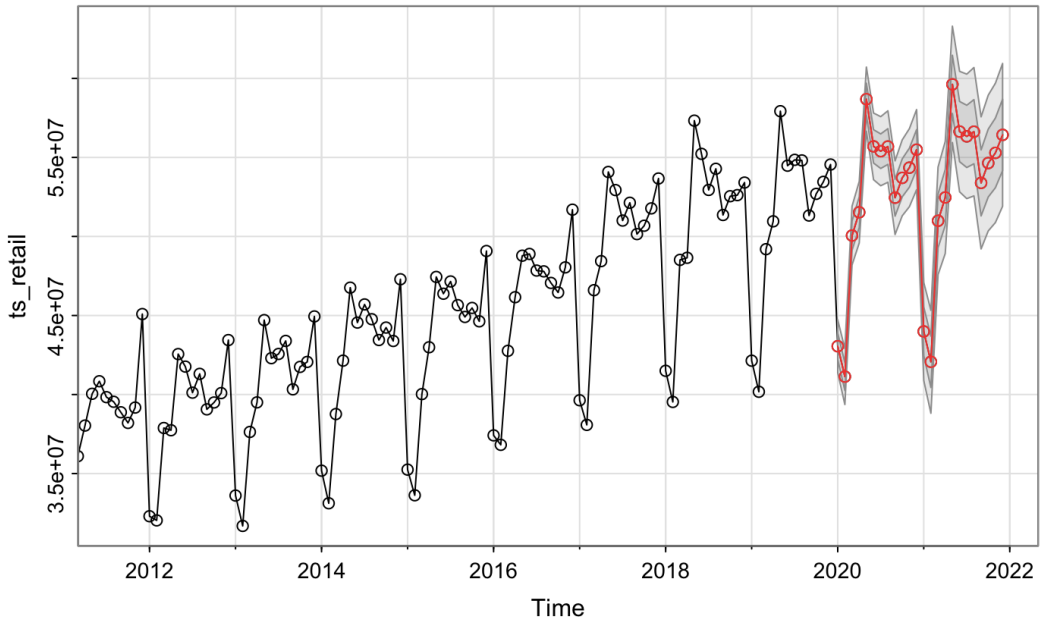


Figure 8: 24-month-ahead forecast of Canadian retail trade sales (January 2020 – December 2021) from the SARIMA$(0,1,1)\times(0,1,1)_{12}$ model fitted on 1992–2019. Shaded regions: 80% (dark) and 95% (light) prediction intervals.

4 Conclusion

This study applied SARIMA modelling to 336 monthly observations of Canadian retail trade sales from 1992 to 2019 to forecast seasonal demand. The selected SARIMA$(0,1,1)\times(0,1,1)_{12}$ model, adequately captured the trend and seasonal structure of the series. During out of sample validation from 2016 to 2019, the model performed well on the test data with low error rate, demonstrating genuine predictive power.

Limitations. This analysis has several limitations. First, SARIMA is a linear model unable to

accommodate nonlinear dynamics or asymmetric responses to economic shocks. Second, the univariate model ignores informative macroeconomic covariates like the Bank of Canada policy rate or unemployment rates. Third, excluding the pandemic period from 2020 onwards limits the immediate operational applicability of the forecasts. Finally, the assumption of constant variance may be violated due to growing seasonal amplitude. A logarithmic transformation could address this in future work.

Future Directions. Future research could explore the ARIMAX model by incorporating exogenous regressors like the Bank of Canada overnight rate to improve accuracy during monetary policy changes. Additionally, regime switching models could help capture the structural break introduced by the pandemic. Finally, neural network autoregression and hybrid statistical models represent promising alternatives for capturing the nonlinear features left in the residuals.

References

Statistics Canada. (2024). *Table 20-10-0008-01: Retail trade sales by industry (x 1,000)*. https://doi.org/10.25318/2010000801-eng